

APPENDIX A

Practicals

The list below gives key words for practicals written in the statistical language S and intended to accompany the chapters of the book. The practicals themselves may be downloaded from

<http://statwww.epfl.ch/people/~davison/SM>

together with a library of functions and data.

2. Variation

1. Speed of light data. Exploratory data analysis.
2. Maths marks data. Brush and spin plots.
3. Probability plots for simulated data.
4. Illustration of central limit theorem using simulated data.
5. Data on air-conditioning failures. Exponential probability plots.

3. Uncertainty

1. Properties of half-normal distribution. Half-normal plot.
2. Simulation of Student t statistic, following original derivation.
3. Simulation of Wiener process and Brownian bridge.
4. Normal random number generation by summing uniform variables.
5. Implementation and assessment of a linear congruential generator.
6. Coverage of Student t confidence interval under various scenarios.

4. Likelihood

1. Loss of information due to rounding of normal data.
2. Birth data. Maximum likelihood estimation for Poisson and gamma models. Assessment of fit.
3. Data on sizes of groups of people. Maximum likelihood fit of truncated Poisson distribution. Pearson's statistic.
4. α -particle data. Maximum likelihood fit of Poisson process model.

5. Blood group data. Maximum likelihood fit of multinomial model.
6. Generalized Pareto distribution. Nonregular estimation of endpoint.

5. Models

1. Boiling point of water data. Straight-line regression.
2. Survival data on leukaemia. Exponential and Weibull models.
3. HUS data. EM algorithm for mixture of Poisson distributions.
4. EM algorithm for mixture of normal distributions.

6. Stochastic Models

1. Markov chain fitting to Alofi rainfall data. Assessment of fit.
2. Multivariate normal fit to data on head sizes.
3. Time series analysis of Manaus river height data. ARMA modelling.
4. Inhomogeneous Poisson process fitted to freezes of Lake Constance.
5. Extreme-value analysis of FTSE return data.

7. Theory

1. Neurological data. Kernel density estimation. Test of unimodality.
2. Mean integrated squared error of kernel density estimator applied to mixtures of normal densities.
3. Test for spatial Poisson process. Beetle data.
4. Coverage of confidence intervals for Poisson mean.

8. Linear Models

1. Cherry tree data. Linear model.
2. Salinity data. Linear model.
3. Data on IQs of identical twins. Linear model.
4. Cement data. Simulation of collinear data.
5. Simulation to assess properties of stepwise model selection procedures.
6. Data on pollution and mortality. Linear model. Ridge regression.

9. Designed Experiments

1. Chick bone data. Inter- and intra-block recovery of information.
2. Millet plant data. Latin square. Outliers. Orthogonal polynomials.
3. Data on marking of examination scripts. Analysis of variance.
4. Teak plant data. 2×3 factorial experiment.

10. Nonlinear Models

1. Space shuttle data. Logistic regression model.
2. Beetle data. Regression models for binary data.
3. Stomach ulcer data. Logistic regression for 2×2 tables. Overdispersion.
4. Speed limit data. Log-linear model. Logistic regression model.
5. Lizards data. Log-linear model. Logistic regression model.
6. Titanic survivor data. Log-linear model.

7. Seed germination data. Overdispersion. Quasi-likelihood. Beta-binomial model.
8. Coal-mining disaster data. Inhomogeneous Poisson process. Generalized additive model.
9. Urine crystal data. Logistic regression.
10. Survival data on leukaemia. Proportional hazards model.
11. Motorette data. Survival data analysis.
12. PBC data. Survival data analysis. Proportional hazards model.

11. Bayesian Models

1. Coin spun on edge. Updating individual and group priors.
2. Cloth data. Hierarchical Poisson model. Laplace approximation.
3. Gibbs sampler for bivariate truncated exponential distribution.
4. Random walk Metropolis–Hastings algorithm with Cauchy proposals.
5. Pump failure data. Gibbs sampler for hierarchical Poisson model.
6. HUS data. Gibbs sampler. Changepoint in Poisson variables.
7. Beaver body temperature data. Gibbs sampler. Changepoint in normal variables.
8. Data augmentation algorithm with multinomial data.

12. Marginal and Conditional Likelihood

1. Ancillary statistic. Simulation with Cauchy data.
2. Saddlepoint approximation. Laplace distribution.
3. Urine data. Logistic regression. Approximate conditional inference.